

HW/Lab 1 Sol: Modulation, Upconversion, BER

ECSE 4560/6560: Modern Communication Systems

RPI Fall 2025

Theory Problems

1. Complex Baseband Representation

The transmitted signal from a radio has the following form

$$\tilde{s}(t) = \text{Re}\{s(t)e^{j\omega_c t}\}$$

where $\tilde{s}(t)$ is the passband signal, $s(t)$ is corresponding baseband signal, and $\omega_c = 2\pi f_c$ is the radian frequency for up-conversion.

Consider the following passband BPSK signal:

$$\tilde{s}(t) = A \cos(\omega_c t + a_i \pi)$$

and QPSK signals

$$\tilde{s}(t) = A \cos(\omega_c t + a_i \pi) - A \sin(\omega_c t + b_i \pi)$$

where A is a constant amplitude and a_i and b_i are independent binary random variables that can take on values $a_i, b_i \in \{0, 1\}$.

- a) Find the complex baseband representation $s(t)$ of the *BPSK signal*.

Solution

The complex-baseband representation is given by:

$$s(t) = A \cos a_i \pi = A(\cos a_i \pi + j \sin a_i \pi) = Ae^{ja_i \pi},$$

since we can then generate $\tilde{s}(t) = A \cos(\omega_c t + a_i \pi) = \text{Re}\{s(t)e^{j\omega_c t}\}$.

- b) Find the complex baseband representation $s(t)$ of the *QPSK signal*.

Solution

The complex-baseband representation is given by:

$$s(t) = A(\cos a_i \pi + j \cos b_i \pi) = Ae^{ja_i \pi} + Ae^{jb_i \pi},$$

since we can then generate $\tilde{s}(t) = A \cos(\omega_c t + a_i \pi) - A \sin(\omega_c t + b_i \pi) = \text{Re}\{s(t)e^{j\omega_c t}\}$.

2. BER, theoretical expression for BPSK, QPSK

Now, take the $s(t)$ defined in the above question. In a practical setting, the received signal, $r(t)$ will include both the transmitted signal and noise.

$$r(t) = s(t) + n(t)$$

Where $n(t)$ can be viewed as a realization of a Gaussian random variable with zero mean and variance σ^2 . Note for complex $n(t)$, we have

$$n(t) = n_I(t) + jn_Q(t), \quad (1)$$

$$n_I(t), n_Q(t) \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \quad \sigma^2 = \frac{N_0}{2}. \quad (2)$$

Assume that $s(t)$ sends bits 0, 1 each in an equi-probable 1/2 probability. Also assume that bit is decoded from $r(t)$ by observing the phase $\angle r(t)$ and declaring bit 0 is $-\frac{\pi}{2} \leq \angle r(t) \leq \frac{\pi}{2}$, and bit 1 otherwise.

- a) Compute conditional probabilities $p_{1|0}$ and $p_{0|1}$. Here $p_{1|0}$ denotes the probability that bit 0 was sent but instead bit 1 is decoded by observing $\angle r(t)$ instead. $p_{0|1}$ likewise denotes the probability that bit 1 was sent but bit 0 is decoded from $\angle r(t)$.

Solution

$$p_{1|0} = \Pr\{n_I(t) > A\} = Q\left(\frac{A}{\sigma}\right) = Q\left(\frac{A}{\sqrt{\frac{N_0}{2}}}\right) = Q\left(\sqrt{\frac{2A^2}{N_0}}\right) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right),$$

$$p_{0|1} = \Pr\{n_I(t) < -A\} = Q\left(\frac{A}{\sigma}\right) = Q\left(\frac{A}{\sqrt{\frac{N_0}{2}}}\right) = Q\left(\sqrt{\frac{2A^2}{N_0}}\right) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right).$$

- b) Use this to compute probability of error: $p_e = \frac{1}{2}p_{1|0} + \frac{1}{2}p_{0|1}$.

Solution

$$p_e = \frac{1}{2}p_{1|0} + \frac{1}{2}p_{0|1} = Q\left(\sqrt{\frac{2A^2}{N_0}}\right) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right).$$

- c) (ECSE 6560 Only) Provide a sketch solution to compute p_e for QPSK. Why are p_e for QPSK and BPSK the same?

Solution

Follow the same steps as the BPSK bit error rate for the real part signal and imaginary part signal, respectively. The bit error rates are identical because BPSK and QPSK here have identical magnitude A for real and imaginary part signals.

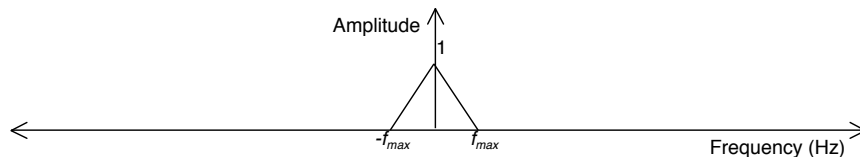
3. Spectrum of AM modulated signals

Let $s(t)$ be a bandlimited baseband signal with a frequency content $S(f)$ given by $1 - |f|/f_{\max}$, where f is the frequency and $f_{\max}$ is the maximum frequency of the baseband signal. This baseband signal is multiplied by $\cos(2\pi f_c t)$ to produce an amplitude-modulated passband signal $\tilde{s}(t) = s(t) \cos(2\pi f_c t)$, where f_c is the carrier frequency and $f_c \gg f_{\max}$.

- a) Sketch the amplitude of the frequency spectrum of the baseband signal $S(f)$.

Solution

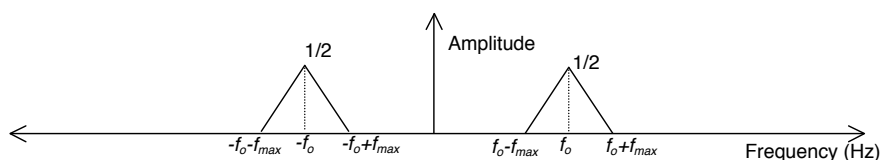
The amplitude of the baseband frequency spectrum is just a triangular function.



- b) Sketch the amplitude of the frequency spectrum of the amplitude-modulated passband signal $\tilde{S}(f)$.

Solution

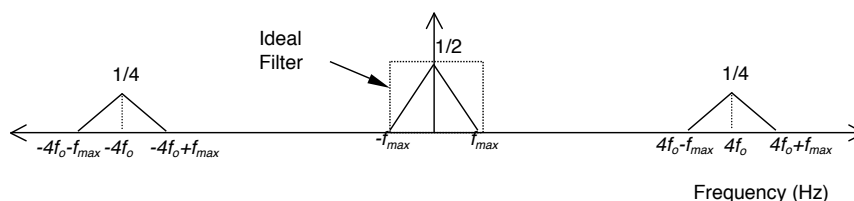
The amplitude of the modulated spectrum is.



- c) The AM signal is down-converted by multiplying $\tilde{s}(t)$ by a coherent signal of the form $\cos(2\pi f_c t)$. The signal is filtered by an ideal lowpass filter with a cutoff frequency $f_{\max}$. Sketch the amplitude of the frequency spectrum of the down-converted signal.

Solution

Applying the modulation property of Fourier transform again yields

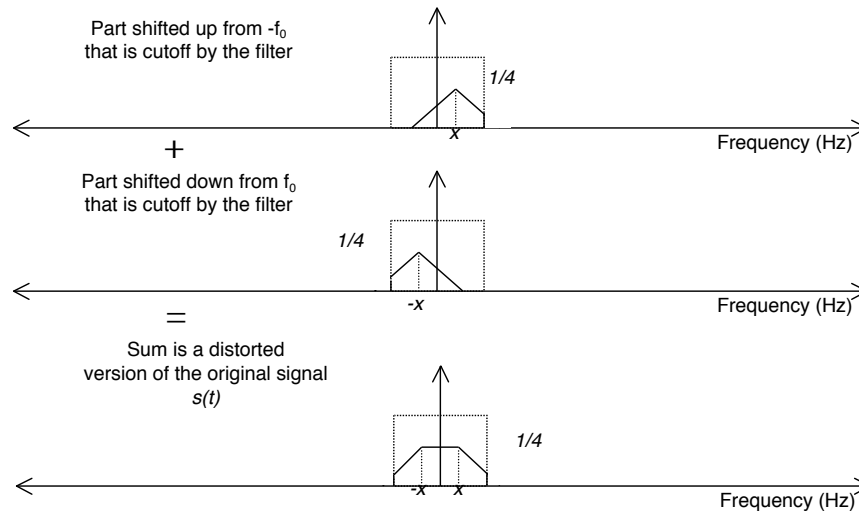


- d) *Frequency Down-conversion errors*

Here we consider the effect of a frequency error in the down-conversion. An AM signal $\tilde{s}(t) = s(t) \cos(2\pi f_c t)$ is down-converted using $\cos(2\pi f_c [1 + x]t)$ where x is a relative frequency error. Sketch the amplitude of the frequency spectrum of the down-converted signal for: (a) $x = 0$, (b) $x = f_{\max}/(10f_c)$, and (c) $x = f_{\max}/f_c$. Comment on the results. Does a frequency down-conversion error cause distortion at Nyquist rate of the baseband signal? Explain.

Solution

When $x \neq 0$, the two parts of the spectrum do not exactly overlap. This is shown schematically below



When the offset $x = f_{\max}/f_c$ then the baseband spectrum is ‘inverted’. The distortion is caused by the two parts of the spectrum not lining up due to the different demodulation frequency used and the filter which cuts out the higher frequency components of the baseband signal. The Nyquist rate of the baseband signal does not change because the filter cuts out all higher frequency components. If you have assumed that no such filter exists, then the Nyquist rate will increase. Either answer backed by valid assumptions will be given points.

Matlab Simulations

Spectral Analysis using the FFT, Power Spectrum

```

1  close all;
2  clear all;
3
4  %% Problem 1
5
6  nFFT = 32;
7  n = 0:nFFT-1;
8  fs = 8192;
9  fc = 256;
10 Ts = 1/fs;
11
12 %%
13 %
14 % $$x(n)=Acos(2\pi f_{\{c\}}nT_{\{s\}})$$
15 %
16 x = cos(2*pi*fc*n*Ts);
17 %%

```

```

18
19 % a.) Using the generated time-series x(n), apply the 32-point FFT
20
21 X = fft(x);
22
23 % b.) Generate the power time series of X[k] denoted P[k] = (X[k])
24
25 P = abs(X).^2;
26
27 % c.) plot P[k]
28
29 figure;
30 subplot(3,1,1);
31 stem(P);
32 xlabel('Sample Number');
33
34 % d.)
35 Pshifted = fftshift(P);
36 subplot(3,1,2);
37 stem(Pshifted);
38 xlabel('Sample Number');
39
40 % e.)
41 % i.) the freq spacing is fs/N, where N is the fft size.
42 delF = fs/nFFT;
43
44 F = ((-nFFT/2):(nFFT/2-1))*(fs/nFFT);
45 subplot(3,1,3);
46 stem(F,Pshifted);
47 xlim([-fs/2 fs/2]);
48 xlabel('Frequency (Hz)');
49
50 % f.) plot in db
51 figure;
52 stem(F,db(Pshifted)/2);
53 xlim([-fs/2 fs/2]);
54 ylim([-40 40]);
55 xlabel('Frequency (Hz)');
56
57 %% Problem 2
58
59 L = 128;
60 N = 8;
61 nFFT = 1024;
62 fs = 1;
63
64 F = linspace(-0.5,0.5-(1/nFFT),nFFT);
65

```

```

66 % a.)
67
68 % bpsk
69 x1 = kron(randsrc(1,L),ones(1,N));
70 % ask
71 x2 = (x1 + 1)/2;
72 % awgn
73 x3 = randn(1,L*N);
74
75 figure;
76 subplot(3,1,1)
77 plot(F, fftshift(db(abs(fft(x1)))));
78 ylim([-20 60]);
79 title('BPSK')
80 xlabel('Normalized Frequency')
81
82 subplot(3,1,2)
83 plot(F, fftshift(db(abs(fft(x2)))));
84 ylim([-20 60]);
85 title('ASK')
86 xlabel('Normalized Frequency')
87
88 subplot(3,1,3)
89 plot(F, fftshift(db(abs(fft(x3)))));
90 ylim([-20 60]);
91 title('AWGN')
92 xlabel('Normalized Frequency')

```

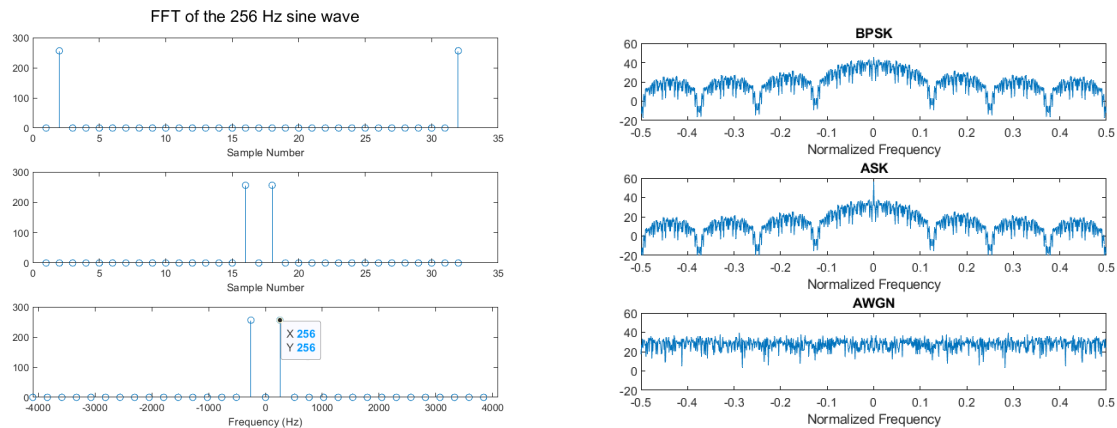


Figure 1: (a) We can observe the peaks at 256 Hz and determine the frequency of sine wave via fft analysis (b) Power spectrum of BPSK, ASK and AWGN

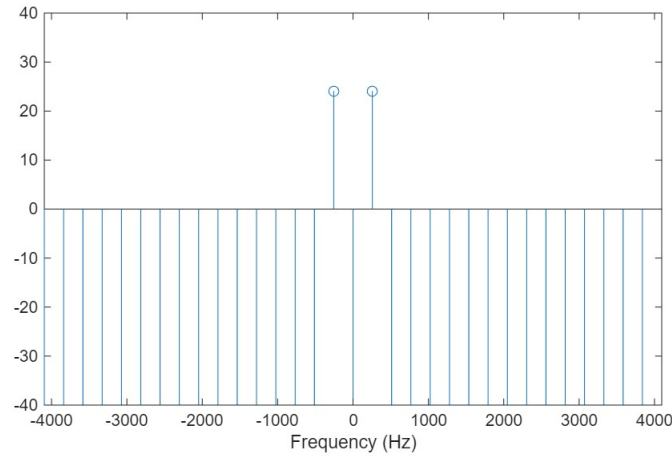


Figure 2: Output of 1.(f)

We observe that BPSK and ASK have sinc function like behaviour. The difference between ASK and BPSK is that BPSK does not have a high DC content whereas ASK has high DC (at 0 zero frequency) as it has a change in amplitude. This change in amplitude creates a step effect which is like square wave with modulation. BPSK does not have a change in amplitude and only a phase change, so it has lesser DC content. The AWGN is random and hence has equal contributions from all the frequency components. It is called Additive White Gaussian Noise and the "white" here refers to the fact that similar to white light which contains all the colors in equal proportion, white noise contains all the frequencies in equal proportion.